Ch. 7 Non-Textbook Questions

Principles of Microeconomics

July 16, 2026

1. Consider a market where supply is given by $Q_S = 10P - 40$ and demand is given by $Q_D = 200 - 10P$.

   (a) What is the market equilibrium price and quantity?

   (b) What is consumer surplus at the market equilibrium?

   (c) What is producer surplus at the market equilibrium?

   (d) If the government instead implements a price ceiling of \$9, what quantity of the good gets traded?

   (e) In the world of the price ceiling, what are consumer surplus and producer surplus?

   (f) In the world of the price ceiling, what is deadweight loss?

   (g) Who benefits and who loses as a result of the price ceiling?

2. The corn lobby convinces Congress to pass a price floor on corn to boost corn farmers' earnings. Suppose demand for corn is $Q_D = 60 - 10P$ and supply is $Q_S = 5P$.

   (a) If Congress implements a price floor of \$4, how does producer surplus change?

   (b) If Congress implements a price floor of \$5 instead, how does producer surplus change?

   (c) Who benefits and who loses as a result of the \$5 price floor?

3. Demand for gasoline is given by $Q_D = 105 - 15P$. Supply is $Q_S = 25P - 15$. To reduce greenhouse gas emissions, Congress passes a tax on gasoline. By law, producers have to pay \$1.60 in taxes for every gallon sold.

   (a) What quantity of gasoline is sold after the tax?

   (b) What price do consumers pay, and what price do producers receive after the tax?

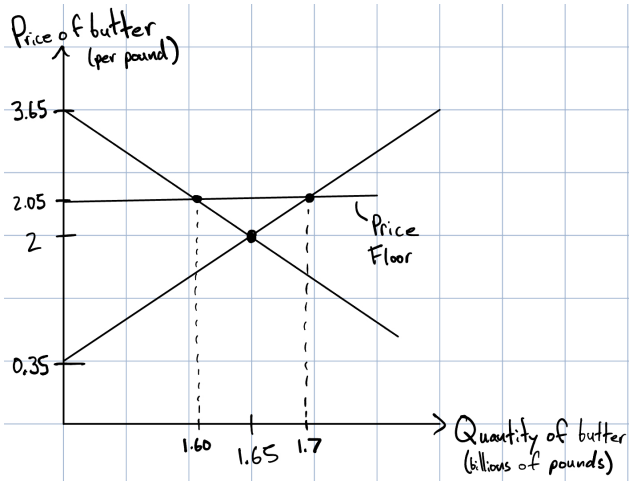
   (c) What are consumer and producer surplus after the tax?

   (d) What's government revenue after the tax?

   (e) What's deadweight loss after the tax?

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4. The U.S. Department of Agriculture (USDA) administers a price floor for butter which the 2008 Farm Bill set at $2.05 per pound. At that price, the quantity of butter supplied in 2010 was 1.7 billion pounds. The quantity demanded was 1.6 billion pounds. To support dairy farmers, the USDA chose to buy up the 100 million pound surplus. The accompanying diagram illustrates the market for butter in 2010.



(a) In the absence of a price floor, how much consumer surplus would there have been? How much producer surplus? What would total surplus have been?

(b) What was consumer surplus with the price floor?

(c) What was producer surplus with the price floor? Recall that producers sold 1.7 billion pounds of butter – some to consumers and some to the USDA.

(d) How much money did the USDA spend buying up surplus butter?

(e) The USDA had to spend tax revenue to buy up the surplus butter. Therefore, to calculate total surplus for the market, we have to subtract the decrease in tax revenue. Using your answers from parts (b)-(d), what was total surplus in the market for butter? How does this compare to total surplus without a price floor?